15 (a) Factorise $a^2 - b^2$

.....

(1)

(b) Hence, or otherwise, simplify fully $(x^2 + 4)^2 - (x^2 - 2)^2$

.....

(3)

(Total for Question 15 is 4 marks)

16 There are only red counters, blue counters and purple counters in a bag.
The ratio of the number of red counters to the number of blue counters is 3 : 17

Sam takes at random a counter from the bag.
The probability that the counter is purple is 0.2

Work out the probability that Sam takes a red counter.

P(red or blue) = 1 – 0.2 = 0.8

Share 0.8 between 3 + 17 = 20 ratio units.

Red	Blue	Sum
3	17	20
×0.04 ↓	×0.04 ↓	×0.04 ↓
0.12	0.68	0.8

1 mark
0.8 ÷ 20 = 0.04 per unit

1 mark
3 × 0.04 for red

P(red) = 0.12

1 mark
Probability in decimal notation 0.12

(Total for Question 16 is 3 marks)

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